

```
!pip install nltk scikit-learn matplotlib
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)
Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (2.0.2)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.1)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

```
import nltk
import string
import matplotlib.pyplot as plt

from nltk.corpus import movie_reviews, stopwords
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
nltk.download('movie_reviews')
nltk.download('stopwords')
nltk.download('punkt')
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package movie_reviews to /root/nltk_data...
[nltk_data] Package movie_reviews is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
True
```

```
positive_reviews = []
negative_reviews = []

for fileid in movie_reviews.fileids('pos'):
    positive_reviews.append(movie_reviews.raw(fileid))

for fileid in movie_reviews.fileids('neg'):
    negative_reviews.append(movie_reviews.raw(fileid))
```

```
stop_words = set(stopwords.words('english'))

def preprocess(text):
    tokens = nltk.word_tokenize(text.lower())
    words = [
        word for word in tokens
        if word.isalpha() and word not in stop_words
    ]
    return " ".join(words)
```

```
positive_clean = [preprocess(text) for text in positive_reviews]
negative_clean = [preprocess(text) for text in negative_reviews]
```

```
print(len(positive_clean))
print(len(negative_clean))
```

```
1000
1000
```

```
print(positive_clean[0][:500])
```

```
films adapted comic books plenty success whether superheroes batman superman spawn geared toward kids casper arthouse crowd
```

```
tfidf_pos = TfidfVectorizer(max_features=1000)
tfidf_neg = TfidfVectorizer(max_features=1000)

pos_tfidf_matrix = tfidf_pos.fit_transform(positive_clean)
neg_tfidf_matrix = tfidf_neg.fit_transform(negative_clean)
```

```
import numpy as np

pos_scores = np.asarray(pos_tfidf_matrix.mean(axis=0)).ravel()
pos_terms = tfidf_pos.get_feature_names_out()

pos_tfidf = dict(zip(pos_terms, pos_scores))
top_pos = sorted(pos_tfidf.items(), key=lambda x: x[1], reverse=True)[:15]

print("Top 15 Positive TF-IDF Words:")
for word, score in top_pos:
    print(word, ":", round(score, 4))
```

```
Top 15 Positive TF-IDF Words:
film : 0.0998
movie : 0.0633
one : 0.059
like : 0.0412
story : 0.0347
good : 0.0331
life : 0.0325
also : 0.0322
time : 0.0313
character : 0.0306
even : 0.0305
characters : 0.03
would : 0.0298
two : 0.0289
well : 0.0288
```

```
neg_scores = np.asarray(neg_tfidf_matrix.mean(axis=0)).ravel()
neg_terms = tfidf_neg.get_feature_names_out()

neg_tfidf = dict(zip(neg_terms, neg_scores))
top_neg = sorted(neg_tfidf.items(), key=lambda x: x[1], reverse=True)[:15]

print("\nTop 15 Negative TF-IDF Words:")
for word, score in top_neg:
    print(word, ":", round(score, 4))
```

```
Top 15 Negative TF-IDF Words:
film : 0.0945
movie : 0.0772
one : 0.0593
like : 0.0459
even : 0.0373
would : 0.0346
good : 0.0341
bad : 0.0337
time : 0.0327
story : 0.0323
get : 0.0317
much : 0.031
character : 0.0301
characters : 0.0297
could : 0.0295
```

```
pos_words, pos_values = zip(*top_pos)
neg_words, neg_values = zip(*top_neg)

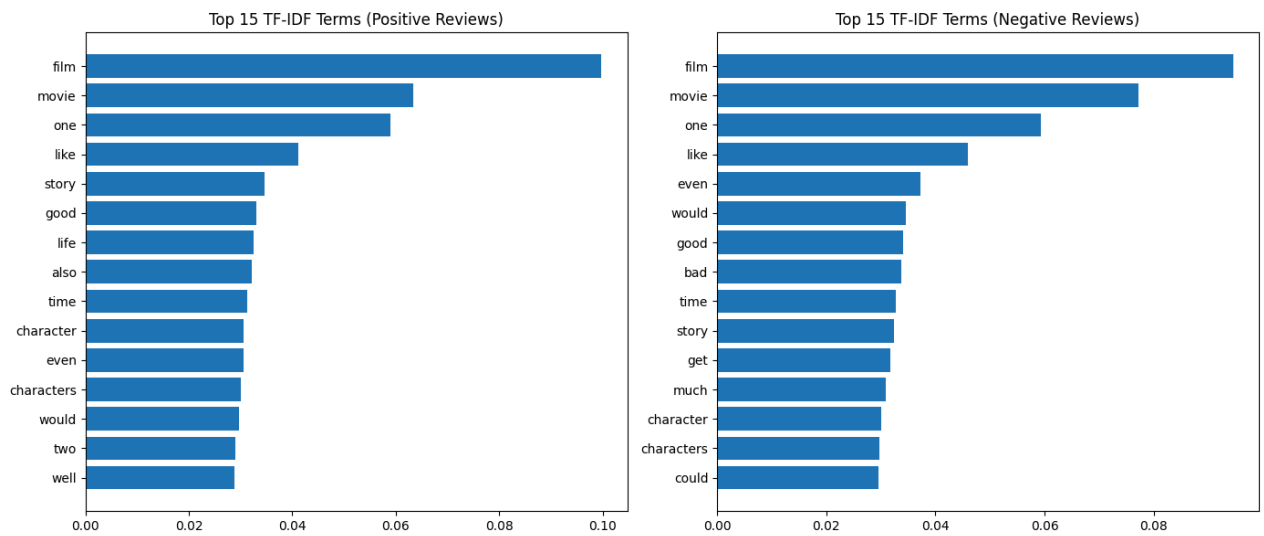
plt.figure(figsize=(14,6))

plt.subplot(1,2,1)
plt.barh(pos_words, pos_values)
plt.title("Top 15 TF-IDF Terms (Positive Reviews)")
```

```
plt.gca().invert_yaxis()

plt.subplot(1,2,2)
plt.barh(neg_words, neg_values)
plt.title("Top 15 TF-IDF Terms (Negative Reviews)")
plt.gca().invert_yaxis()

plt.tight_layout()
plt.show()
```



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